CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 462 Solutions

Midterm 462 · 25 marks · a graded student paper

Muharib

01 Solution Q1: linear probing

Key	key mod 7	Probes	Cell
76	6	6	**6**
93	2	2	**2**
40	5	5	**5**
47	5	5 → 6 → 0	**0**
10	3	3	**3**
55	6	6 → 0 → 1	**1**

0	1	2	3	4	5	6
47	55	93	10		40	76

```
47:  h = 47 mod 7 = 5 taken   →   6 taken   →   (6 + 1) mod 7 = 0
```

```
55:  h = 55 mod 7 = 6 taken   →   0 taken   →   1
```

This is the student's own answer, which received 5 out of 5. Note the wrap-around from the end of the table (6) to its start (0).

02 Solution Q2: True / False + MCQ

#	Answer	Reason
1	**a. True**	A character that occurs more often gets a shorter code.
2	**b. False**	Greedy does not give the optimum for every DP problem.
3	**c**	Dynamic programming. The student chose it and it was marked correct, even though the wording here says "future results" instead of "past results".
4	**b. False**	The correct equation ends with `T(MERGE)`, not $T(\frac{MERGE}{2})$.
5	**c. Other value**	The time is `O(n)` when the activities are sorted by finish time. The student chose b and it was marked wrong.
6	**a. True**	DP solves each sub-problem once and stores its result, which saves a lot of time. The student chose False and it was marked wrong.
7	**d**	Little-omega means strictly greater. Greater than or equal is big-Omega.
8	**C. O(n²)**	Two nested loops, each running `n` times.
9	**A. O(1)**	The best case: the element is in the first cell. The student chose `O(n)` and it was marked wrong.
10	**b**	Binary search halves the range every time.

The student scored 2.5 out of 5, answering items 1, 2, 3, 4 and 8 correctly. The correct answers to the other five items come from the marker's highlights on the paper.

03 Solution Q3: greedy knapsack

Greedy sorts the items by value per kilogram (value ÷ weight) from largest to smallest and takes them in that order until the bag is full.

Item	Weight (kg)	Value (SAR)	Value per kg	Taken
Saffron	2	4000	2000	2 kg → 4000
Cardamom	2	1000	500	2 kg → 1000
Cinnamon	5	500	100	1 kg → 100
Pepper	10	500	50	0

```
Weight used = 2 + 2 + 1 = 5 kg
```

```
Total value = 4000 + 1000 + 100 = 5100 SAR
```

Where is the problem? If items must be taken whole (0-1 knapsack), greedy takes saffron and cardamom and stops at 4 kg and SAR 5000, leaving one kilogram it cannot fill. The fix is to allow taking part of an item (fractional knapsack), so we take one kilogram of cinnamon and the total becomes 5100.

The student scored 2.5 out of 5: the marker crossed out his first total and ticked his last line "4000 + 1000 + 100 = 5100 SAR" and the fractional knapsack idea.

04 Solution Q4: LCS

```
X[i] = Y[j]:   c[i, j] = c[i−1, j−1] + 1        diagonal + 1
```

```
X[i] ≠ Y[j]:   c[i, j] = max(top, left)
```

	∅	A	B	C	A	D	F
∅	0	0	0	0	0	0	0
A	0	**1**	1	1	1	1	1
C	0	1	1	**2**	2	2	2
B	0	1	2	2	2	2	2
A	0	1	2	2	**3**	3	3
E	0	1	2	2	3	3	3
D	0	1	2	2	3	**4**	4

```
Length = 4
```

```
LCS = ACAD        (ABAD is also a correct answer)
```

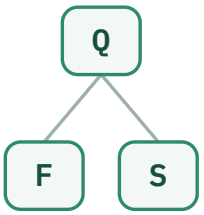
The student wrote `ACAD` and received 5 out of 5. The coloured cells are the traceback path from the last cell: at every match we take the letter and move diagonally.

05 Solution Q5: B-tree

A B-tree of degree 3: every node has at most 3 children, which means at most 2 keys. When a node reaches 3 keys it splits: the middle key moves up to the parent, and the first and third become separate nodes.

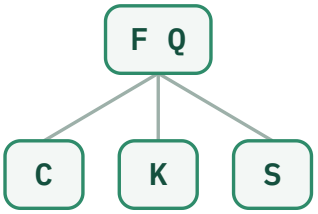
1 F, S, Q

`F S`, then Q makes it `F Q S`: three keys, so it splits and Q moves up.



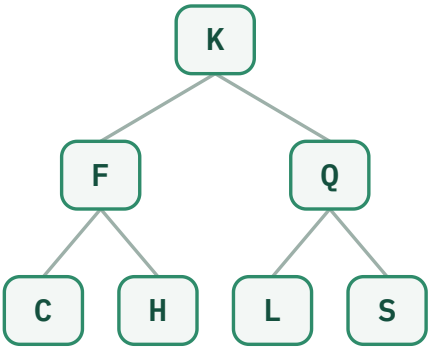
2 K, C

`F K`, then C makes it `C F K`, so it splits and F moves up to the root.

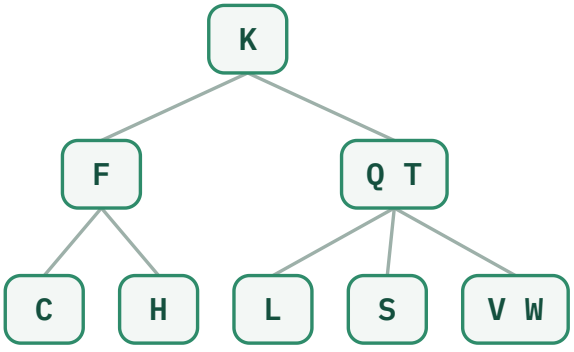


3 L, H

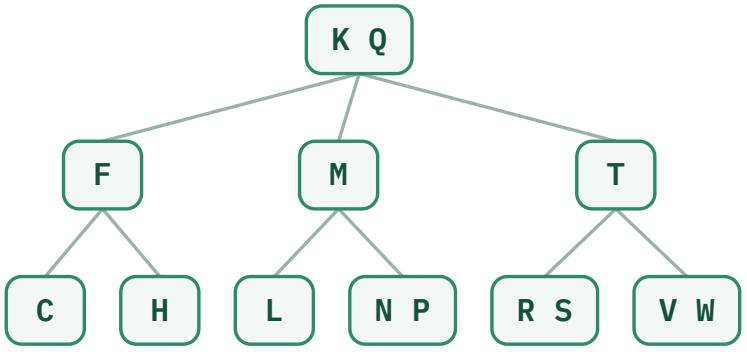
`K L`, then H makes it `H K L`: K moves up, so the root becomes `F K Q` and splits too, and K becomes the new root.



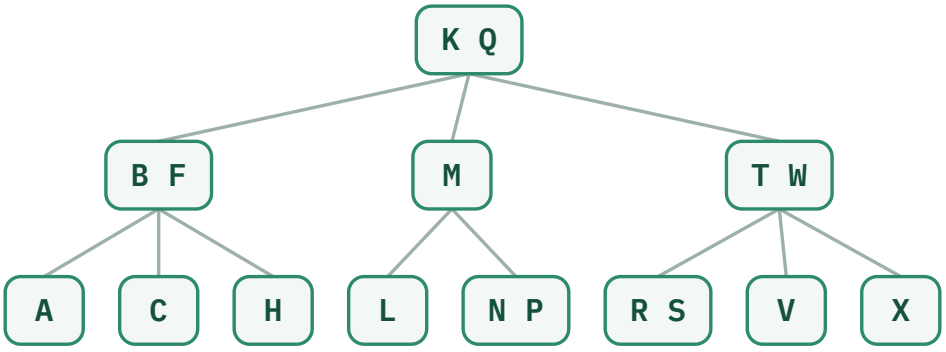
4 T, V, W



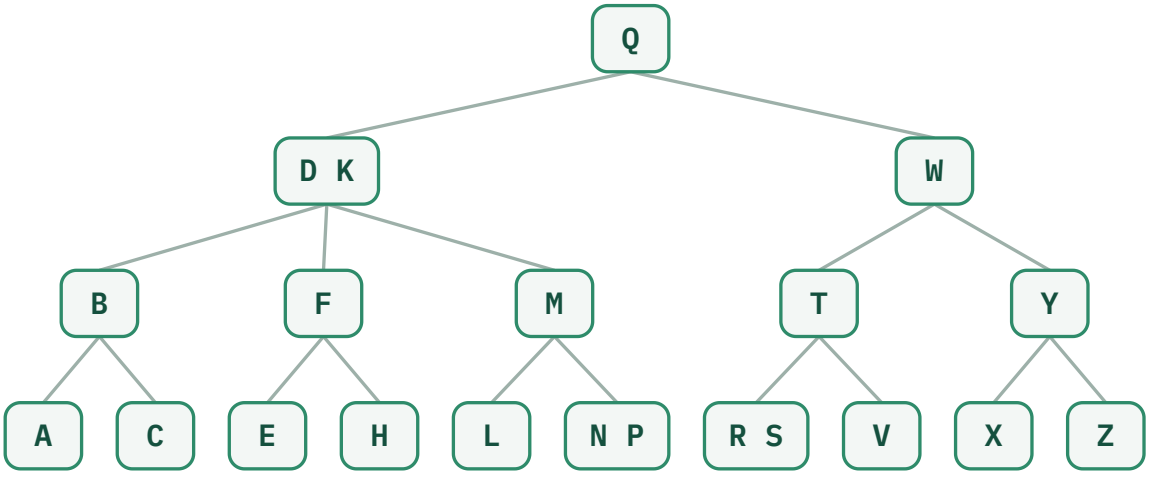
5 M, R, N, P



6 A, B, X



7 Y, D, Z, E · the final tree



The final tree matches the empty tree given in the question exactly: one root, two nodes under it, five under those, and ten leaves. The student scored 4.25 out of 5 because he wrote `F K` instead of `D K` and got two leaves wrong.